UNIVERSITI TEKNOLOGI MARA
FINAL EXAMINATION

COURSE	INTRODUCTION TO INTERACTIVE MULTIMEDIA
COURSE CODE	CSC413
EXAMINATION	NOVEMBER 2026
TIME	3 HOURS

INSTRUCTIONS TO CANDIDATES

1. This question paper consists of the questions printed in this booklet.
2. Answer ALL questions in the Answer Booklet. Start each answer on a new page.
3. Do not bring any material into the examination room unless permission is given by the invigilator.
4. Please check to make sure that this examination pack consists of:
   i) the Question Paper
   ii) an Answer Booklet - provided by the Faculty
5. Answer ALL questions in English.

DO NOT TURN THIS PAGE UNTIL YOU ARE TOLD TO DO SO

This examination paper consists of 6 printed pages.

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PART A (15 MARKS)

State TRUE (T) or FALSE (F) for the following statements.

1. Path animation moves an object along a predefined route.
2. A prototype is an early model of a product.
3. RGB is an additive colour model.
4. Sampling converts a continuous sound wave into digital values.
5. A linear presentation allows users to choose their own path.
6. Kerning adjusts the space between individual character pairs.
7. A web browser displays hypertext documents.
8. Card-based authoring tools are best for messages with a clear beginning and end.
9. Hertz measures sound frequency.
10. Sans serif fonts are generally preferred for screen display.
11. Tweening creates intermediate frames between keyframes.
12. Streaming requires the whole file to download before playback.
13. HTML is used to structure content on web pages.
14. Higher resolution always means a smaller file.
15. Chroma key compositing replaces a colour background with another image.

PART B (15 MARKS)

Select the most appropriate answer from Table A for each of the following questions.

Table A. Matching options

A. Multimedia	B. Bitmap	C. Linear multimedia
D. Card-based tool	E. Browser	F. Scanner
G. Vector	H. Keyframe	I. Sampling
J. Chroma key	K. Storyboard	L. MIDI
M. Tweening	N. Frame rate	O. Target audience

1. important frame in an animation
2. input device that digitises documents
3. visual sequence plan
4. converting sound into digital values
5. software that displays web pages
6. image made of pixels
7. standard for musical notes and instructions
8. frames displayed per second
9. integration of text, graphics, audio, video and animation
10. presentation without user control of sequence
11. replacing a colour background
12. intended users of a product
13. authoring tool organised like pages or cards
14. image made of mathematical objects
15. generating in-between frames

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PART C (30 MARKS)

Choose the correct answer for each question.

1. Which is an input device?
   A. Printer
   B. Plotter
   C. Scanner
   D. Monitor

2. Convergence in multimedia refers to:
   A. Printing images
   B. Recording audio only
   C. Deleting media files
   D. Merging of media and computing

3. Streaming allows media to:
   A. Remove compression
   B. Play while downloading
   C. Require full download first
   D. Only play offline

4. A serif font has:
   A. Only uppercase letters
   B. No spacing
   C. No strokes
   D. Small strokes at character ends

5. Which is NOT a multimedia element?
   A. Text
   B. Audio
   C. Compiler
   D. Animation

6. A presentation where the user controls the sequence is called:
   A. Linear
   B. Non-linear
   C. Static
   D. Printed

7. A bitmap image is made of:
   A. Mathematical objects
   B. Sound samples
   C. Text characters
   D. Pixels

8. Chroma key is used to:
   A. Add text
   B. Scan images
   C. Replace a colour background
   D. Compress audio

9. RAM is:
   A. Permanent storage
   B. A web browser
   C. An input device
   D. Temporary working memory

10. Which is an authoring tool category?
    A. RGB
    B. Card-based
    C. CMYK
    D. JPEG

11. A prototype is:
    A. A font
    B. A codec
    C. A final release
    D. An early model of a product

12. Anti-aliasing is used to:
    A. Increase file size only
    B. Compress video
    C. Add sound
    D. Smooth jagged edges

13. Keyframes are:
    A. Colour profiles
    B. Audio samples
    C. Web links
    D. Important frames in an animation

14. Which is a benefit of multimedia in education?
A. It combines visuals and sound
B. It uses only text
C. It replaces all teachers
D. It removes all interaction

15. Copyright protects:
A. File names
B. Computer hardware
C. Internet speed
D. Original creative works

16. MIDI stores:
A. Images
B. Musical notes and instructions
C. Video frames
D. Digitized audio samples

17. A hyperlink links to:
A. Only text
B. Another resource
C. Only audio
D. Only images

18. Interactivity in multimedia means:
A. There is no navigation
B. The system plays automatically
C. Only text is shown
D. The user can control content and flow

19. Plug and play means:
A. No power needed
B. Automatic device recognition
C. Wireless only
D. Manual driver installation only

20. Virtual reality creates:
A. A simulated environment
B. A printed page
C. A text font
D. A camera lens

21. Sampling converts:
A. Digital video to text
B. Images to vectors
C. Analog sound to digital values
D. Text to speech

22. Tweening creates:
A. In-between frames
B. Text fonts
C. Sound effects
D. Video codecs

23. Which best defines multimedia?
A. Text, graphics, audio, video and animation
B. Text only
C. A printed book
D. A single image

24. Which is an output device?
A. Keyboard
B. Scanner
C. Mouse
D. Plotter

25. Which colour model is subtractive?
A. CMYK
B. HSL
C. RGB
D. HSV

26. Hypermedia is best described as:
A. A printed manual
B. A single video file
C. Text-only media
D. Multimedia with hyperlinks

27. Which image type scales without losing quality?
A. Vector
B. Bitmap
C. Photograph
D. Scan

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28. Morphing changes:

A. One shape into another
B. Font size
C. File format
D. Audio pitch

29. A graphics tablet is used for:

A. Printing
B. Web browsing
C. Drawing input
D. Sound output

30. An authoring tool is used to:

A. Only browse the web
B. Only record audio
C. Assemble multimedia elements
D. Only print text

PART D (40 MARKS)

QUESTION 1

A shopping complex requires a kiosk that gives customers location and retailer information.

a) State TWO important project team roles for completing the kiosk.

(3 marks)

b) Design a low-fidelity prototype for a pre-school courseware home page. The page must show topic menus, the course title and the school logo.

(3 marks)

c) Discuss TWO factors that lead to the wide use of mobile devices in games and shopping applications.

(4 marks)

QUESTION 2

Text is an essential element in a multimedia project. Its shape, size, design, colour, arrangement and linking all affect the message.

a) State TWO reasons sans serif is mostly used for screen display and serif is used for printed material.

(3 marks)

b) Describe TWO roles of hypertext in multimedia.

(3 marks)

c) Explain how text can have a massive impact on an informative multimedia project in the link and HTML context.

(4 marks)

QUESTION 3

A company wants an interactive product demonstration that runs on a kiosk and on the web.

a) State TWO features to consider when choosing authoring software.

(2 marks)

b) Explain the interactivity level suitable for the demonstration.

(3 marks)

c) Explain TWO authoring tool categories appropriate for this project.

(5 marks)

QUESTION 4

A museum wants a website that presents its collection and lets visitors explore related exhibits.

a) State TWO differences between linear and non-linear navigation.

(2 marks)

b) Describe TWO roles of hyperlinks in the website.

(4 marks)

c) Explain how streaming and compression affect multimedia delivery on the web.

(4 marks)

END OF QUESTION PAPER

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